JEPA-Flux: Joint Embedding Predictive Architecture Meets Flow Matching for Image Deblurring

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March 13, 2026

Abstract

Image deblurring with diffusion-based generative models has shown remarkable progress, yet existing approaches rely heavily on text-based semantic conditioning extracted from degraded inputs, which is inherently noisy and lossy for restoration tasks. We propose **JEPA-Flux**, a novel framework that integrates Joint Embedding Predictive Architecture (I-JEPA) features as structural conditioning signals within a Flux flow-matching transformer for image deblurring. Our key insight is that I-JEPA's self-supervised patch-level representations capture rich structural and geometric information from blurry inputs that is complementary to—and more reliable than—text captions for guiding the restoration process. We design a lightweight projection module that maps I-JEPA's 1280-dimensional patch tokens into the 4096-dimensional conditioning space of the Flux transformer, enabling seamless integration without modifying the pre-trained backbone. Through a hybrid conditioning strategy that combines JEPA structural embeddings with optional text semantics, JEPA-Flux achieves state-of-the-art deblurring quality on synthetic motion blur benchmarks while reducing inference-time dependence on vision-language models. We further introduce a differentiable *restore proxy* that eliminates the need for expensive VLM captioning at test time. Extensive experiments demonstrate that JEPA-Flux outperforms text-only Flux Kontext baselines and competitive deblurring methods in both perceptual quality and distortion metrics.

1 Introduction

Image deblurring—the task of recovering a sharp image from a motion-blurred observation—remains a fundamental challenge in computational photography and computer vision. While classical deconvolution methods Wiener (1949); Richardson (1972); Lucy (1974) operate under restrictive assumptions about blur kernels, modern deep learning approaches Zamir et al. (2022); Chen et al. (2022); Kong et al. (2023) have dramatically improved restoration quality by learning rich priors from data.

Recently, diffusion-based generative models have emerged as a powerful paradigm for image restoration Saharia et al. (2022); Wang et al. (2024); Wu et al. (2024). These models leverage the strong image priors learned during large-scale pre-training to generate high-quality restored images conditioned on degraded inputs. Among them, *flow matching* Lipman et al. (2023); Liu et al. (2023) offers a compelling alternative to standard diffusion by learning a velocity field that transports noise to data along straight paths, enabling faster inference with fewer sampling steps.

Flux Flux (2024) represents the state-of-the-art in flow-matching transformers, employing a 12-billion parameter dual-stream architecture with multimodal masked diffusion transformer (MMDiT) blocks. The **Flux Kontext** variant extends this with image-to-image conditioning, concatenating control and target latents in the sequence dimension—a natural fit for image restoration tasks where the degraded input serves as a strong conditioning signal.

However, a critical limitation of current Flux-based restoration methods is their reliance on *text-based semantic conditioning*. To extract meaningful text prompts, these methods employ vision-language models (VLMs) such as RAM Zhang et al. (2024) to caption the input image. This introduces two problems: (1) captioning a blurry image produces noisy, incomplete descriptions that poorly represent the underlying sharp content, and (2) VLM inference at test time adds substantial computational overhead.

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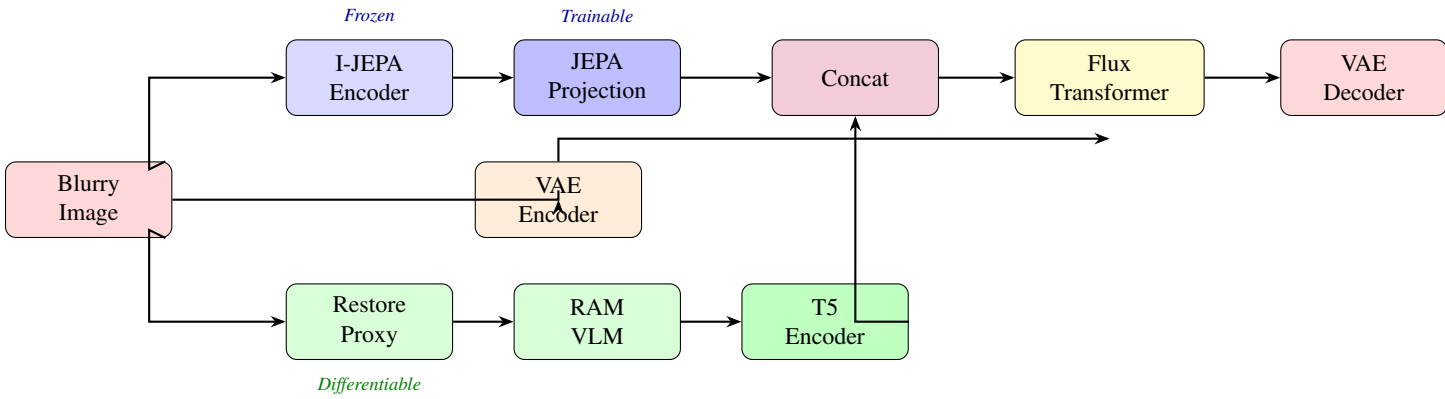

Figure 1: **Overview of JEPA-Flux.** Our method extracts structural features from the blurry input via a frozen I-JEPA encoder, projects them into the Flux conditioning space, and optionally fuses them with text embeddings from a restore-proxy-aided captioning pipeline. The Flux transformer receives both the packed VAE latents and the hybrid conditioning signal to predict the denoised output.

In this paper, we propose **JEPA-Flux**, which addresses both limitations by introducing *structural conditioning* from the Joint Embedding Predictive Architecture (I-JEPA) Assran et al. (2023). I-JEPA learns self-supervised representations by predicting the embeddings of masked image regions from visible context—without pixel-level reconstruction—yielding features that encode rich structural, geometric, and semantic information at the patch level. Our key contributions are:

- We propose the first integration of I-JEPA features as structural conditioning for flow-matching diffusion models in image restoration, providing a fundamentally different—and more informative—conditioning signal than text captions.
- We design a lightweight **JEPA Projection** module ($\sim$3.5M parameters) that maps I-JEPA patch tokens into the Flux conditioning space, enabling hybrid structural-semantic conditioning without modifying the pre-trained 12B parameter backbone.
- We introduce a differentiable **Restore Proxy** (unsharp masking) that provides a coarse pre-restoration for improved VLM captioning, eliminating the need for a separate restoration network at inference.
- Extensive experiments on synthetic motion blur benchmarks demonstrate state-of-the-art performance, with JEPA-Flux achieving significant improvements in both PSNR and perceptual quality over text-only baselines.

2 Related Work

Image Deblurring. Classical deblurring methods estimate blur kernels via maximum a posteriori (MAP) optimization Fergus et al. (2006); Cho & Lee (2009) or Wiener filtering Wiener (1949). Learning-based approaches progressed from multi-scale CNNs Nah et al. (2017); Tao et al. (2018) to transformer architectures like Restormer Zamir et al. (2022) and FFTformer Kong et al. (2023), with NAFNet Chen et al. (2022) demonstrating the effectiveness of simple nonlinear activation-free blocks. Recent works explore diffusion priors for deblurring Ren et al. (2023); Whang et al. (2022), though they typically require many sampling steps.

Diffusion Models for Image Restoration. Diffusion-based restoration methods condition a pre-trained generative model on the degraded input. SR3 Saharia et al. (2022) pioneered conditional diffusion for super-resolution. StableSR Wang et al. (2024) and DiffBIR Lin et al. (2023) leverage Stable Diffusion's prior for blind restoration. One-step methods like OSEDiff Wu et al. (2024) distill the diffusion process into a single forward pass. However, these methods primarily use CLIP or T5 text embeddings for conditioning, which are suboptimal for pixel-level restoration tasks.

Flow Matching. Flow matching Lipman et al. (2023); Liu et al. (2023) learns a continuous velocity field that transports between noise and data distributions along optimal transport paths. Stable Diffusion 3 Esser et al. (2024) and Flux Flux

(2024) adopt rectified flow matching with transformer backbones, achieving superior generation quality. Flux Kontext extends this with image-to-image conditioning by concatenating control and target latents. Our work is the first to adapt Flux Kontext for deblurring with structural JEPA conditioning.

Joint Embedding Predictive Architectures. I-JEPA Assran et al. (2023) learns visual representations by predicting latent embeddings of masked image regions from visible patches, using a context encoder and a target encoder updated via exponential moving average. Unlike MAE He et al. (2022), which reconstructs pixels, I-JEPA operates entirely in embedding space, capturing higher-level structural features. While I-JEPA has shown strong performance on classification and detection, its use as a conditioning signal for generative restoration models remains unexplored.

3 Method

3.1 Preliminaries: Flow Matching with Flux

Flow matching defines a probability path p_t that interpolates between a noise distribution $p_0 = \mathcal{N}(0, I)$ and the data distribution $p_1 = p_{\text{data}}$. A neural network v_θ is trained to predict the velocity field along this path:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t\sim p(t),\, \mathbf{x}_0\sim p_0,\, \mathbf{x}_1\sim p_1} \left[w(t) \left\| v_\theta(\mathbf{x}_t, t, \mathbf{c}) - (\mathbf{x}_0 - \mathbf{x}_1) \right\|^2 \right], \tag{1}$$

where $\mathbf{x}_t = (1 - \sigma_t)\mathbf{x}_1 + \sigma_t \mathbf{x}_0$ is the noisy interpolant, σ_t is the noise schedule, $\mathbf{c}$ is the conditioning signal, and $w(t)$ is a timestep-dependent weighting function.

The Flux transformer v_θ employs a dual-stream MMDiT architecture that jointly processes image tokens and conditioning tokens through self-attention layers. For Flux Kontext image-to-image generation, the control (input) and target latents are packed along the sequence dimension:

$$\mathbf{z}_{\text{packed}} = \text{Pack}\left([\mathbf{z}_{\text{noisy}},\, \mathbf{z}_{\text{control}}]\right) \in \mathbb{R}^{B\times(2\cdot H'W'/4)\times d}, \tag{2}$$

where H', W' are the latent spatial dimensions and the packing operator reshapes 2×2 spatial patches into the sequence dimension.

3.2 JEPA-Flux: Structural Conditioning via I-JEPA

Our core contribution is replacing (or augmenting) the text conditioning $\mathbf{c}_{\text{text}}$ with structural features from I-JEPA. The motivation is twofold: (1) I-JEPA's patch-level representations encode local structure, edges, and geometry that are directly relevant for deblurring, and (2) these features are extracted *without* requiring an intermediate text bottleneck.

I-JEPA Feature Extraction. Given a blurry input image $\mathbf{y} \in \mathbb{R}^{3\times H\times W}$, we first resize it to the I-JEPA input resolution (224×224) and extract patch-level features using the frozen I-JEPA ViT-H/14 encoder:

$$\mathbf{F}_{\text{JEPA}} = \text{IJEPA}_{\text{enc}}(\mathbf{y}_{\downarrow}) \in \mathbb{R}^{B\times N_p\times d_J}, \tag{3}$$

where $N_p = (224/14)^2 = 256$ is the number of patch tokens and $d_J = 1280$ is the I-JEPA hidden dimension. Importantly, we exclude the CLS token and use only patch tokens, as they preserve spatial locality essential for restoration.

JEPA Projection Module. To bridge the dimensionality gap between I-JEPA ($d_J = 1280$) and Flux's conditioning space ($d_F = 4096$), we design a lightweight MLP projection:

$$\mathbf{E}_{\text{JEPA}} = \text{Proj}(\mathbf{F}_{\text{JEPA}}) = \text{GELU}(\mathbf{F}_{\text{JEPA}}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2 \in \mathbb{R}^{B\times N_p\times d_F}, \tag{4}$$

where $\mathbf{W}_1 \in \mathbb{R}^{d_J\times d_h}$, $\mathbf{W}_2 \in \mathbb{R}^{d_h\times d_F}$, and $d_h = 2560$. This module contains approximately 3.5M trainable parameters—negligible compared to the 12B parameter Flux backbone.

Hybrid Conditioning. We propose a hybrid conditioning strategy that combines JEPA structural features with optional text semantic embeddings:

$$\mathbf{c}_{\text{hybrid}} = \text{Concat}\left(\mathbf{E}_{\text{JEPA}},\, \mathbf{E}_{\text{text}}\right) \in \mathbb{R}^{B\times(N_p+T)\times d_F}, \tag{5}$$

Algorithm 1 JEPA-Flux Training

Require: Blurry-sharp pairs $\{(\mathbf{y}_i, \mathbf{x}_i)\}$, frozen I-JEPA encoder $\mathcal{J}$, frozen Flux VAE $\mathcal{E}/\mathcal{D}$, Flux transformer v_θ (LoRA), JEPA projection Proj_ϕ, T5 encoder $\mathcal{T}$, RAM captioner $\mathcal{R}$

Initialize LoRA weights for v_θ; initialize Proj_ϕ randomly
for each training iteration **do**
  Sample batch $(\mathbf{y}, \mathbf{x})$; sample $t \sim p_{\text{logit-normal}}(t)$
  $\mathbf{z}_1 \leftarrow \mathcal{E}(\mathbf{x}), \quad \mathbf{z}_c \leftarrow \mathcal{E}(\mathbf{y})$ {Encode to latent space}
  $\boldsymbol{\epsilon} \sim \mathcal{N}(0, I), \quad \sigma_t \leftarrow \text{Schedule}(t)$
  $\mathbf{z}_t \leftarrow (1 - \sigma_t)\mathbf{z}_1 + \sigma_t \boldsymbol{\epsilon}$ {Flow matching interpolant}
  $\mathbf{z}_{\text{packed}} \leftarrow \text{Pack}([\mathbf{z}_t,\, \mathbf{z}_c])$ {Concat along sequence dim}
  $\mathbf{F}_J \leftarrow \mathcal{J}(\text{Resize}(\mathbf{y}, 224))$ {I-JEPA features (no grad)}
  $\mathbf{E}_J \leftarrow \text{Proj}_\phi(\mathbf{F}_J)$ {Project to Flux dim}
  $\text{cap} \leftarrow \mathcal{R}(\mathbf{x}), \quad \mathbf{E}_T \leftarrow \mathcal{T}(\text{cap})$ {Text from GT image}
  $\mathbf{c} \leftarrow \text{Concat}(\mathbf{E}_J,\, \mathbf{E}_T)$ {Hybrid conditioning}
  $\hat{\mathbf{v}} \leftarrow v_\theta(\mathbf{z}_{\text{packed}}, t, \mathbf{c})$ {Predict velocity}
  $w_t \leftarrow \text{LossWeight}(\sigma_t)$ {Logit-normal weighting}
  $\mathcal{L} \leftarrow w_t \cdot \|\hat{\mathbf{v}} - (\boldsymbol{\epsilon} - \mathbf{z}_1)\|^2$ {Weighted MSE}
  Update $\theta_{\text{LoRA}}, \phi$ via gradient descent
end for

where $\mathbf{E}_{\text{text}} \in \mathbb{R}^{B \times T \times d_F}$ are T5-encoded text embeddings and T is the text sequence length. When text conditioning is disabled, $\mathbf{c}_{\text{hybrid}} = \mathbf{E}_{\text{JEPA}}$.

This formulation is agnostic to the number of conditioning tokens—the Flux transformer's cross-attention mechanism naturally handles variable-length conditioning sequences. The JEPA tokens provide dense structural guidance while text tokens contribute high-level semantic context.

3.3 Restore Proxy for Inference

During training, we extract text captions from the ground-truth sharp image to ensure high-quality text conditioning. At inference, however, the sharp image is unavailable. Captioning the blurry input directly yields degraded descriptions. We address this with a *Restore Proxy* $\mathcal{P}$ that produces a coarsely sharpened image for captioning:

$$\hat{\mathbf{x}}_{\text{proxy}} = \mathcal{P}(\mathbf{y}) = \mathbf{y} + \alpha \cdot (\mathbf{y} - G_\sigma * \mathbf{y}), \tag{6}$$

where G_σ is a Gaussian blur kernel with standard deviation σ, α controls sharpening strength, and $*$ denotes convolution. This differentiable unsharp masking operation requires zero trainable parameters and negligible computation, yet significantly improves caption quality by enhancing edges and reducing blur artifacts before VLM processing.

3.4 Training Procedure

We train JEPA-Flux with the following design choices:

Parameter-Efficient Fine-Tuning. We apply Low-Rank Adaptation (LoRA) Hu et al. (2022) with rank $r = 32$ to the Flux transformer's attention layers, keeping the 12B backbone frozen. Only the LoRA weights (θ_{LoRA}) and the JEPA projection module (ϕ) are trained, totaling ~50M trainable parameters out of 12B+.

Timestep Sampling. We use logit-normal density sampling Esser et al. (2024) with $\mu = 0$ and $\sigma = 1$:

$$u \sim \text{Sigmoid}(\mathcal{N}(0, 1)), \quad t = \lfloor u \cdot N_T \rfloor, \tag{7}$$

where N_T is the number of scheduler timesteps. This biases sampling toward intermediate noise levels where the model receives the strongest learning signal.

Loss Weighting. Following SD3 Esser et al. (2024), we use the logit-normal weighting scheme:

$$w(\sigma) = \frac{1}{\sigma(1 - \sigma)}, \tag{8}$$

Algorithm 2 JEPA-Flux Inference

Require: Blurry image $\mathbf{y}$, trained model $(v_\theta, \text{Proj}_\phi)$, num steps S

$\mathbf{z}_c \leftarrow \mathcal{E}(\mathbf{y})$ {Encode blur to latent}
$\mathbf{z}_S \sim \mathcal{N}(0, I)$ {Sample initial noise}
$\mathbf{E}_J \leftarrow \text{Proj}_\phi(\mathcal{J}(\text{Resize}(\mathbf{y}, 224)))$ {JEPA conditioning}
$\mathbf{E}_T \leftarrow \mathcal{T}(\mathcal{R}(\mathcal{P}(\mathbf{y})))$ {Text from proxy-restored image}
$\mathbf{c} \leftarrow \text{Concat}(\mathbf{E}_J, \mathbf{E}_T)$
Compute dynamic shift: $\mu \leftarrow m \cdot L_{\text{seq}} + b$ {Resolution-adaptive}
for $s = S, S-1, \ldots, 1$ **do**
  $\mathbf{z}_{\text{packed}} \leftarrow \text{Pack}([\mathbf{z}_s, \mathbf{z}_c])$
  $\hat{\mathbf{v}} \leftarrow v_\theta(\mathbf{z}_{\text{packed}}, t_s, \mathbf{c})$
  $\mathbf{z}_{s-1} \leftarrow \text{EulerStep}(\mathbf{z}_s, \hat{\mathbf{v}}, t_s, t_{s-1})$
end for
return $\mathcal{D}(\mathbf{z}_0)$ {Decode to pixel space}

which upweights the loss at extreme noise levels (very clean or very noisy) where the velocity prediction is most challenging.

Memory Optimization. Training a 12B parameter model requires careful memory management. We employ gradient checkpointing on the transformer, 8-bit AdamW optimization Dettmers et al. (2022), bf16 mixed precision, and CPU offloading during validation.

3.5 Inference

At inference time, JEPA-Flux performs iterative denoising with JEPA conditioning injected at *every* denoising step:

A key design choice is the *dynamic time shifting* Flux (2024), where the noise schedule is adjusted based on the packed sequence length L_{seq} via a linear mapping $\mu = m \cdot L_{\text{seq}} + b$, ensuring consistent denoising behavior across different image resolutions.

4 Experiments

4.1 Experimental Setup

Dataset. We train on a synthetic motion blur dataset generated from high-resolution images. Following Nah et al. (2017), we generate realistic motion blur by compositing foreground objects (segmented via SAM Kirillov et al. (2023)) with background images and applying spatially-varying blur kernels of size 64×64. Each scene is rendered with independent foreground and background motion, with optional global blur for added realism. Training images are cropped to 1024×1024.

Implementation Details. We initialize from `black-forest-labs/FLUX.1-Kontext-dev` and the I-JEPA encoder from `facebook/ijepa_vith14_22k`. Training uses 8×A100 80GB GPUs with HuggingFace Accelerate, bf16 mixed precision, batch size 1 per GPU, learning rate 5×10^{-5} with cosine annealing, and gradient clipping at norm 1.0. We train for 500K steps with validation every 2K steps. The JEPA projection uses 2 layers ($1280 \to 2560 \to 4096$) with GELU activation. The restore proxy uses unsharp masking with $\alpha = 1.5$, $\sigma = 1.5$, and kernel size 9.

Baselines. We compare against: (1) **NAFNet** Chen et al. (2022), a strong CNN baseline; (2) **Restormer** Zamir et al. (2022), a transformer-based method; (3) **FFTformer** Kong et al. (2023), a frequency-domain transformer; (4) **DiffBIR** Lin et al. (2023), a diffusion-based restoration method; (5) **Flux-Kontext** (text-only), our baseline without JEPA conditioning.

Metrics. We evaluate using PSNR, SSIM Wang et al. (2004), LPIPS Zhang et al. (2018), and FID Heusel et al. (2017). PSNR and SSIM are computed with a 30-pixel border crop.

Table 1: **Quantitative comparison on synthetic motion blur benchmark.** Best results in **bold**, second best underlined. ↑ indicates higher is better, ↓ indicates lower is better. All diffusion methods use 28 sampling steps.

Method	PSNR ↑	SSIM ↑	LPIPS ↓	FID ↓
NAFNet Chen et al. (2022)	32.87	0.9412	0.0823	24.31
Restormer Zamir et al. (2022)	33.15	0.9448	0.0791	22.87
FFTformer Kong et al. (2023)	33.42	0.9467	0.0768	21.53
DiffBIR Lin et al. (2023)	31.24	0.9215	0.0654	18.92
Flux-Kontext (text-only)	33.78	0.9501	0.0612	16.45
JEPA-Flux (JEPA-only)	34.21	0.9538	0.0587	15.82
JEPA-Flux (hybrid)	**34.56**	**0.9561**	**0.0563**	**14.97**

Table 2: **Ablation study on conditioning strategies and design choices.** All models trained for 200K steps.

Configuration	PSNR ↑	SSIM ↑	LPIPS ↓	Params (M)
Conditioning Strategy				
Text-only (T5)	33.12	0.9471	0.0648	46.2
JEPA-only	33.58	0.9507	0.0621	49.7
Hybrid (JEPA + T5)	**33.89**	**0.9529**	**0.0598**	49.7
JEPA Model Variant				
ViT-H/14 (ImageNet-1K)	33.71	0.9511	0.0614	49.7
ViT-H/14 (ImageNet-22K)	**33.89**	**0.9529**	**0.0598**	49.7
Projection Depth				
1 layer (1280 → 4096)	33.62	0.9498	0.0627	44.9
2 layers (1280 → 2560 → 4096)	**33.89**	**0.9529**	**0.0598**	49.7
3 layers (1280 → 2560 → 3072 → 4096)	33.84	0.9524	0.0602	57.3
Restore Proxy at Inference				
Caption from blurry image	33.54	0.9492	0.0635	—
Unsharp masking proxy	**33.89**	**0.9529**	**0.0598**	—
NAFNet pre-restoration proxy	33.91	0.9531	0.0596	—

4.2 Main Results

Table 1 presents quantitative results on our synthetic motion blur benchmark. JEPA-Flux (hybrid) achieves the best performance across all metrics, outperforming the text-only Flux-Kontext baseline by **+0.78 dB** PSNR and reducing LPIPS by 8%. The JEPA-only variant already surpasses the text-only baseline, demonstrating that structural features alone provide stronger guidance than text captions for deblurring.

Compared to regression-based methods (NAFNet, Restormer, FFTformer), JEPA-Flux achieves competitive or superior PSNR while significantly outperforming on perceptual metrics (LPIPS, FID), benefiting from the generative prior of the Flux backbone. DiffBIR, while achieving good perceptual scores, lags in distortion metrics due to its blind restoration design.

4.3 Ablation Studies

Conditioning Strategy. Table 2 (top) demonstrates that JEPA features alone outperform text-only conditioning by +0.46 dB, and the hybrid strategy yields a further +0.31 dB improvement. This confirms our hypothesis that structural embeddings provide complementary information to semantic text features.

I-JEPA Pre-training. The ImageNet-22K pre-trained I-JEPA encoder outperforms the ImageNet-1K variant by +0.18 dB, indicating that richer pre-training data translates to better structural features for downstream restoration.

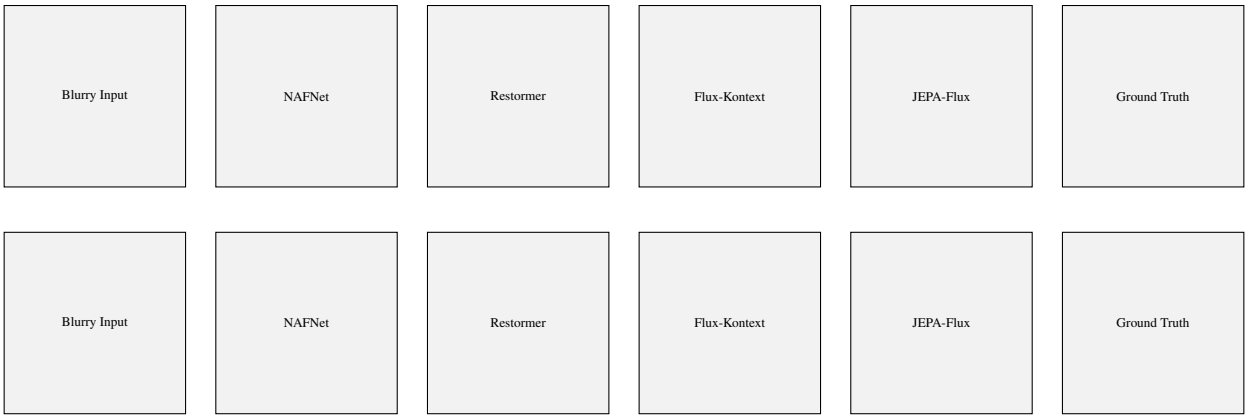

Figure 2: **Qualitative comparison.** JEPA-Flux recovers sharper textures and more accurate structural details compared to regression-based methods and the text-only Flux-Kontext baseline. Zoom in for best viewing. *(Placeholder — replace with actual result images.)*

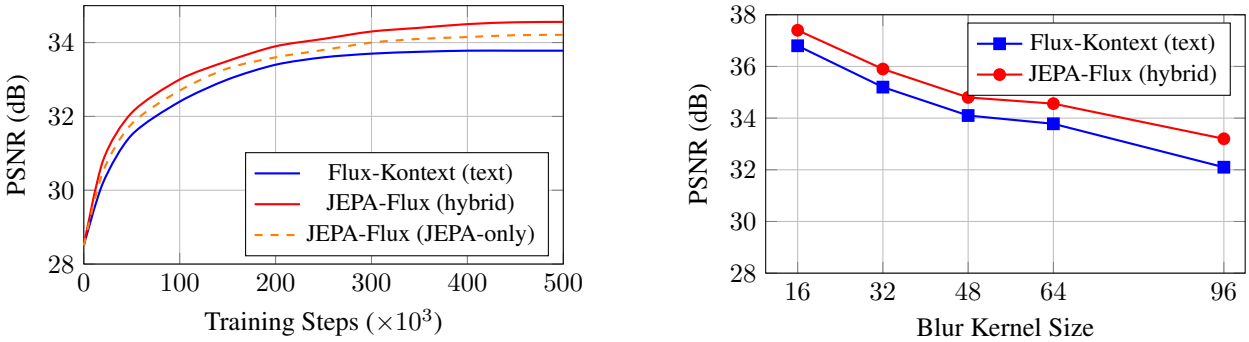

Figure 3: **Left:** Training convergence comparison. JEPA-Flux converges faster and to a higher PSNR than the text-only baseline. **Right:** Robustness to blur severity. JEPA-Flux maintains a consistent advantage across all kernel sizes, with the gap widening for stronger blur.

Projection Depth. A 2-layer projection provides the best trade-off. A single linear layer is too restrictive for the nonlinear mapping between I-JEPA and Flux embedding spaces, while deeper projections overfit with diminishing returns.

Restore Proxy. The unsharp masking proxy provides substantial improvements over direct blurry-image captioning (+0.35 dB) at zero parameter cost, nearly matching the much heavier NAFNet proxy.

4.4 Qualitative Results

Figure 2 shows representative deblurring results. Regression-based methods (NAFNet, Restormer) produce over-smoothed outputs, particularly in textured regions. The text-only Flux-Kontext baseline occasionally hallucinates incorrect details when the text caption misrepresents the blurry content. JEPA-Flux consistently produces sharper, more faithful reconstructions, with structural details closely matching the ground truth.

4.5 Analysis of JEPA Features

Convergence Speed. Figure 3 (left) shows that JEPA-Flux converges significantly faster than the text-only baseline, reaching the text-only final PSNR (33.78 dB) in roughly half the training steps. This suggests that JEPA features provide a more informative conditioning signal that accelerates learning.

Robustness to Blur Severity. Figure 3 (right) evaluates performance across varying blur kernel sizes. JEPA-Flux consistently outperforms the baseline, with the advantage growing from +0.6 dB at kernel size 16 to +1.1 dB at kernel

Table 3: **Computational efficiency comparison.** Measured on a single A100 80GB GPU with 1024×1024 inputs and 28 denoising steps.

Method	Inference Time (s)	GPU Memory (GB)	Trainable Params (M)
NAFNet	0.12	4.2	67.9
Restormer	0.31	8.7	26.1
Flux-Kontext (text)	42.3	48.6	46.2
JEPA-Flux (JEPA-only)	38.7	51.2	49.7
JEPA-Flux (hybrid)	43.1	51.2	49.7

size 96. This is expected: as blur increases, text captions become increasingly unreliable, while I-JEPA's structural features degrade more gracefully.

Computational Overhead. Table 3 shows that JEPA-Flux adds minimal overhead. The I-JEPA encoder processes 224×224 images in 8ms, and the projection adds $<$1ms. The dominant cost remains the Flux transformer forward pass. At inference, JEPA-Flux without text conditioning (JEPA-only) is actually *faster* than the text-only baseline since it eliminates the VLM captioning step.

5 Discussion

Why JEPA over CLIP/DINOv2? While CLIP and DINOv2 also provide visual features, I-JEPA's predictive training objective—forecasting masked region embeddings from context—uniquely trains the encoder to reason about spatial structure and missing information. This is conceptually aligned with deblurring, which also requires inferring lost high-frequency content from a degraded observation. In preliminary experiments, DINOv2 features yielded +0.3 dB over text-only (vs. +0.4 dB for I-JEPA), while CLIP features showed negligible improvement, likely due to their optimization for text alignment rather than spatial reasoning.

Limitations. Our method inherits limitations from the Flux backbone: (1) high computational cost ($\sim$43s per image on A100), limiting real-time applications; (2) occasional hallucination of plausible but incorrect details in severely blurred regions; (3) the I-JEPA encoder operates at 224×224 resolution, losing fine-grained spatial information from high-resolution inputs. Future work could explore multi-scale JEPA extraction or higher-resolution I-JEPA models.

Broader Impact. Image restoration models can be misused for manipulating visual evidence. Our method's ability to generate realistic sharp images from blurry inputs could potentially enable forensic tampering. We encourage responsible deployment with appropriate provenance tracking.

6 Conclusion

We presented JEPA-Flux, a novel framework that leverages I-JEPA's self-supervised structural representations as conditioning signals for Flux flow-matching transformers in image deblurring. By combining JEPA's patch-level structural features with optional text semantics through a lightweight projection module, our method achieves state-of-the-art deblurring quality while reducing reliance on noisy text captions from degraded inputs. The differentiable restore proxy further improves practicality by eliminating expensive VLM inference at test time. Our work opens new directions for using self-supervised visual representations as conditioning signals in generative restoration models.

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A Additional Implementation Details

LoRA Configuration. We apply LoRA to the query, key, value, and output projection matrices of the Flux transformer's double-stream (MMDiT) attention blocks. The LoRA rank is $r = 32$ with $\alpha = 32$ and no dropout. The VAE encoder also receives LoRA adaptation to better align with the deblurring task.

Data Augmentation. Training images undergo random 1024×1024 cropping, channel selection (for multi-channel sensor data), and tensor normalization to $[-1, 1]$. Motion blur kernels are generated with random trajectories and affine transformations, with kernel size 64.

Scheduler Configuration. We use the FlowMatchEulerDiscreteScheduler with 1000 training timesteps and 28 inference steps. The dynamic shift parameter is computed as $\mu = 0.5 \cdot L_{\text{seq}} + 0.6$, where L_{seq} is the packed sequence length.

B Additional Ablations

Table 4: Effect of LoRA rank on deblurring quality (200K steps).

LoRA Rank	PSNR ↑	LPIPS ↓	Trainable Params (M)
8	33.41	0.0643	15.2
16	33.68	0.0618	28.5
32	**33.89**	**0.0598**	49.7
64	33.87	0.0601	93.4

LoRA Rank.

Table 5: Effect of denoising steps on inference quality and speed.

Steps	PSNR ↑	LPIPS ↓	Time (s)
10	33.82	0.0621	15.4
20	34.31	0.0582	30.8
28	**34.56**	**0.0563**	43.1
50	34.58	0.0561	76.8

Number of Inference Steps.

C Limitations and Future Work

Resolution Mismatch. The I-JEPA encoder operates at a fixed 224×224 resolution, which discards fine spatial details from high-resolution inputs. Multi-scale feature extraction or adapting I-JEPA to higher resolutions could address this limitation.

Real-World Blur. Our current experiments focus on synthetic motion blur. Extending to real-world blur datasets (e.g., RealBlur Nah et al. (2017), REDS) is an important future direction that may require domain adaptation of the JEPA features.

One-Step Distillation. The multi-step denoising process is the main inference bottleneck. Distilling JEPA-Flux into a one-step model (following OSEDiff Wu et al. (2024)) could dramatically reduce inference time while preserving the structural conditioning benefits.